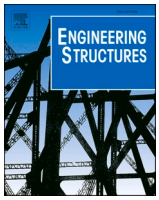
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Convolutional neural networks (CNNs)-based multi-category damage detection and recognition of high-speed rail (HSR) reinforced concrete (RC) bridges using test images

Lingkun Chen [a,b,c,d,*], Wenxin Chen [a], Lu Wang [a], Chencheng Zhai [a], Xiaolun Hu [e], Linlin Sun [f], Yuan Tian [g], Xiaoming Huang [e], Lizhong Jiang [d,h]

[a] College of Architecture Science and Engineering, Yangzhou University, Yangzhou 225127, Jiangsu, China
[b] Department of Civil and Environmental Engineering, University of California, Los Angeles, CA 90095, USA
[c] School of Civil Engineering, Southwest Jiaotong University, Chengdu 610031, Sichuan, China
[d] National Engineering Laboratory for High-speed Railway Construction, Changsha 410004, China
[e] School of Transportation, Southeast University, Nanjing 211189, Jiangsu, China
[f] CCDI(Suzhou) Exploration & Design Consultant Co., LTD. Suzhou, 215123 Jiangsu, China
[g] Transportation Technology Development Promotion Center, China Academy of Transportation Sciences, Beijing 100029, China
[h] School of Civil Engineering, Central South University, Changsha 410075, China

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ABSTRACT

The fast networking of high-speed rail (HSR) may cause in-service fatigue and ultimate load damage to bridges. This paper investigates the application of deep convolutional neural networks (CNNs) for multi-category damage image classification recognition of HSR-reinforced concrete (RC) bridges. The present study establishes a deep learning (DL) system based on a large amount of HSR bridge test data. When to begin, the damage done to HSR bridge piers may be broken down into three primary categories: concrete cracks, concrete spalling, and reinforcement exposure. These categories are determined by the statistics of HSR bridge pier testing. Secondly, in order to develop an automated recognition model for the damage of HSR piers, AlexNet CNNs were taught a transfer learning approach and then used to train themselves. The correct recognition rate of the three damaged pictures in the actual application of the model is 86% for cracks, 82% for reinforcement exposure, and 70% for concrete spalling, all of which have good recognition rates. The study's accuracy and precision enhance detection efficiency and may be utilized to identify HSR pier deterioration quickly. The research comprises a random selection from the training and validation sets. It also assesses the training model's generalization to out-of-sample pictures for engineering applications. This aspect of the work sets it apart from previous research in the same study area.

1. Introduction

In recent years, there have been scientific problems in the field of transportation engineering that need to be solved as soon as possible. These problems include understanding how damage builds over time, how performance degrades during the service life of high-speed rail (HSR) bridges, and finding the best way to maintain, repair, and strengthen them. According to a report by the International Union of Railways (UIC), 11,693 km of HSR railroads are currently being built, which means that the HSR network will grow by almost 25 % in five to six years [1]. Nations that established HSR earlier, such as China, Turkey, Spain, South Korea, and Switzerland, are now perfecting their networks and extending to adjacent countries, which indicates that the development of high-speed rail is continuing at a pretty rapid rate.

In contrast to the escalating construction of railroad bridges, environmental corrosion, material deterioration, and progressive decline of structural performance are more prevalent in highway bridges. According to a 2017 American Society of Civil Engineers survey, 9,1% of

* Corresponding author.
E-mail addresses: lkchen@yzu.edu.cn, lingkunchen08@hotmail.com (L. Chen), MZ120200924@yzu.edu.cn (W. Chen), MZ120200964@yzu.edu.cn (L. Wang), MZ120211019@yzu.edu.cn (C. Zhai), 101010856@seu.edu.cn (X. Hu), 109416327@qq.com (L. Sun), ytian88@163.com (Y. Tian), huangxm@seu.edu.cn (X. Huang), lzhjiang@csu.edu.cn (L. Jiang).

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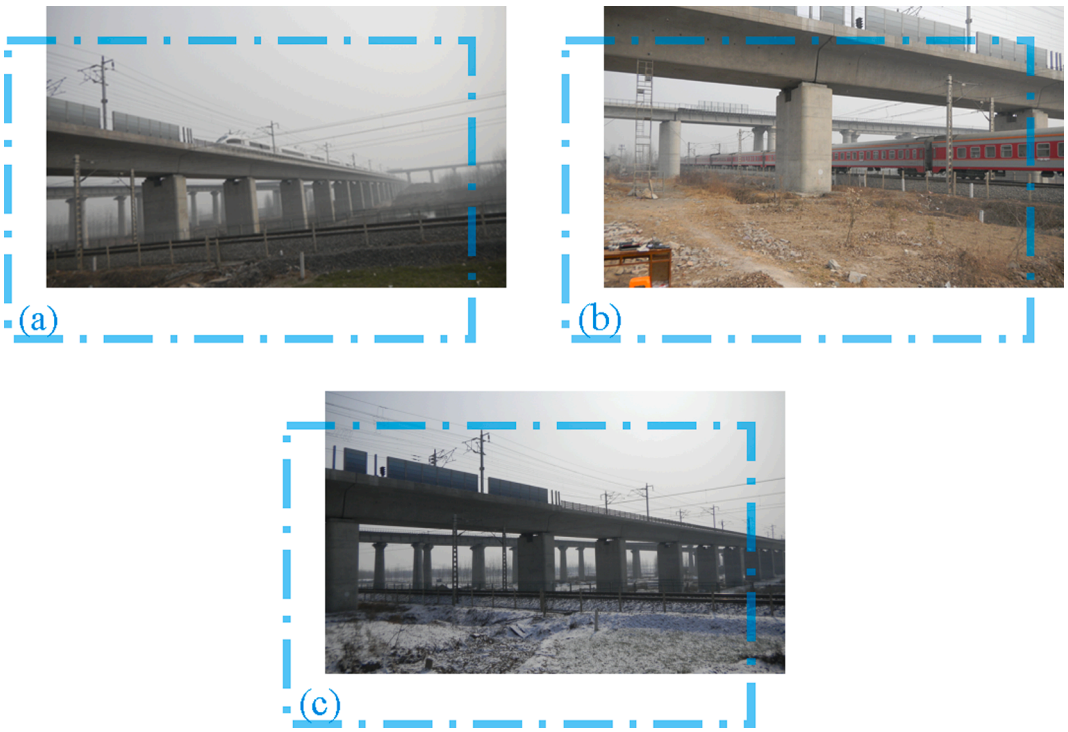


Fig. 1. A comprehensive test for the coupled dynamic performance of the Beijing-Shanghai HSR from 15 to February 17, 2011. (a) Test bridge: Suihe Special Bridge, (b) test site, ordinary trains and HSR trains in separate rail, and (c) test time coinciding with snow, environmental loads interacting.

the nation's 614,387 bridges were structurally inadequate in 2016 [2]. Due to infrastructure growth, the number of bridges in China is growing fast, and so is the need for bridge maintenance, with the number of bridges in need of repair reaching 658,100 [3]. Due to these factors, there is much research on road bridge damage and health monitoring.

In addition to damage induced by in-service loads, complex conditions such as earthquakes have an even greater influence on bridge damage. First, structural damage may result in a loss of the structure's lateral or vertical load-bearing capability [4] or even the complete collapse of the structure, which, in the case of railroad bridges, may cause train derailment incidents on the bridge [5,6]. In addition, the bridge junction bridge structure is a crucial connection in the transportation network, and the earthquake has a cascading impact on the bridge structure.

Post-earthquake damage assessment is essential for rehabilitating and repairing reinforced concrete structures. The evaluation of the status of bridge damage is essential for the quick recovery assessment of transportation networks [7]. In order to automate this procedure and further minimize labor expenses, one of the research hotspots in the subject of bridge inspection is the Investigation of rapid and efficient bridge structural inspection methods and techniques. With the advent of computers and specialized equipment in civil engineering, approaches for the structural health monitoring (SHM) of reinforced concrete structures are used to detect deterioration.

Most SHM methods fall into two groups: those that use vibrations and those that use vision [8]: vibration-based approaches can get information about structural damage in real-time from sensors. The data signals can be watched and changed to find, find, and figure out how much damage has happened. Many studies have focused on time series data, such as acceleration and displacement sensors, and other one-dimensional data in bridge health monitoring. This technique still provides issues in monitoring infrastructure owing to uncertainties and non-uniformly distributed environmental influences, among other considerations. As shown in Fig. 1 for the comprehensive test site of the combined dynamic performance of the Beijing-Shanghai high-speed rail vehicles and bridges from February 15 to 17, 2011, many factors influence the vibration of the surrounding environment if the dynamic response of the bridge is measured.

In particular, the bridge's health monitoring using sensing data is usually only done on essential or critical bridges, and the vast majority of bridges are only evaluated qualitatively. Second, these time-history curves, which seem very helpful, are often not very useful for finding the real state of structural damage [9]. Again, the data collected by the sensors have a lot of "noise," The most important thing for the engineer is to pull out the features and then watch the changes in these features to find the damage [10]. In structural health monitoring, the modal shape and its higher-order derivatives (like the second, third, and fourth-order derivatives) are accurate ways to figure out how much damage there is [11]. It is important to note that the self-oscillation frequency is the most common study, but this has the drawback that it is hard to get good eigenvalues from the natural vibrations of real structures with considerable damage, and most damage to structures is localized. In contrast to the above methods, the "data-driven" method extracts damage-sensitive features by focusing only on the sensor data changes without needing a structural model. Nevertheless, it can take even experienced researchers a long time to test which statistical indices or their derivatives are sensitive to structural damage [12].

Deep learning (DL) has been used in civil engineering in the last few years, especially in SHM, because digital image processing and machine learning algorithms have improved. After an earthquake disaster, drones [13] and robots [14] are used to enter a dangerous environment and take photos of structural damage when people cannot get to the site. However, it is important to know how to get the needed information from the massive bridge damage photos so that earthquake damage predictions can be made quickly and accurately and post-earthquake emergency rescue can be led. Images contain much subtle information, but it was difficult to find until artificial intelligence technology improved. Convolutional neural networks (CNNs) are an important technology in the field of deep learning today. They are good for learning and extracting features from image data [15,16], and they have advantages when learning features of data that look like an image.

Since 2012, Krizhevsky, Sutskever, and Hinton [17] trained a large



Fig. 2. Bridge damage images. (a) Concrete members cracks, (b) exposed reinforcement, (c) spalling, (d) honeycomb, (e) pockmarks,and (f) reinforcement corrosion.

CNN on ImageNet to obtain top-1 and top-5 error rates of 37.5 % and 17.0 % on picture classification, suggesting that CNN-based networks are more accurate in detection. Rapid progress has been made in automated structural damage picture detection, damage identification, and quantitative analysis. Since then, various CNN networks with deeper architectures and backbones have been suggested, including AlexNet, GoogleNet, VGGs, ResNet, Inceptions, Xception, DenseNet, Inception-ResNet, ResNeXt, SqueezeNet, MobileNet, EfficientNet and others [18].

Applications of CNN in bridge engineering classification include the study of concrete crack detection [19], concrete spalling [20], and reinforcement corrosion [21]. Different digital imaging methods for damage detection have been well-studied recently. Gao & Mosalam [22] introduced migration learning (TL) based on VGGNet (Visual Geometry Group) and used and introduced feature extractor and fine-tuning as two TL strategies to apply for structural damage identification. Han et al. [23] used a proposed hybrid technique based on CNNs and digital image processing to identify the crack damage of a member. Based on the You Only Look Once v4 (YOLOv4) network, Cha et al. [24] used the Faster R-CNN architecture to detect five surface damages in concrete and steel structures with an overall recognition accuracy of more than 93 % for 55 test images. Liang et al. [25] used VGG-16 as a migration learning pre-training model for bridge pier damage detection and used Bayesian optimization to verify the robustness of the model. Chun et al. [26] generated sentences describing bridge damage from bridge images by an image captioning method, which could observe multiple forms of bridge damage. Xu et al. [27] proposed an emergency earthquake prediction method based on a long-short-term memory neural network (LSTM). Zhang et al. [28] proposed a machine-learning framework for post-earthquake structural safety assessment. Response and damage models were introduced in this system to enable it to assess the safety of damaged structures after an earthquake quickly. Xiong et al. [29] collected earthquake damage images of building structures with different damage levels from the Internet and trained a CNNs classifier using these images. Using computer vision techniques, Tazarv et al. [30] developed a quantitative definition of the damaged state of reinforced concrete bridge columns after an earthquake.

Predicting bridge pier damage following an earthquake may help restore service or create corrective procedures. Damaged bridges affect post-earthquake emergency transportation and recovery activities. Therefore, early identification is crucial. CNNs for damage identification have mostly concentrated on detecting single cracks in highway bridges; little study has been done on train bridges. Engineers must study railroad bridge damage further.

This work investigates computer vision inspection techniques for structural bridge inspection using a vast amount of HSR bridge test data. This project aims to develop a DL approach that identifies varying degrees of damage to HSR bridge piers, focusing on pier damage detection. Initially, a test database for HSR bridges was established by carefully examining training models and test specimens. Due to the usually inflexible nature of the bridge superstructure, this article focuses on the damage characteristics acquired from the HSR bridge pier tests and separates them into concrete cracks, concrete spalling, and reinforcing exposure. Then, the study integrates migration learning with CNNs to fine-tune the traditional AlexNet CNNs and construct an image identification model for three typical bridge pier damage types. Practical application tests validate the model's applicability and accuracy for recognizing real damage, demonstrating the method's potential utility in engineering applications.

2. Research objective and scope

i) This work focuses on the damage identification of HSR bridge piers; the input pictures are the test photographs.

ii) The authors performed bridge testing between 2013 and 2017; thus, the images featured in this article were shot between 2013 and 2017; cameras: Fuji FinePix S4850, 16 megapixels, and Nikon D3300, 24.78 megapixels;

iii) According to the bridge test findings' statistics, the bridge piers' structural damage was categorized into three primary categories: concrete cracks, concrete spalling, and exposed reinforcement. Since the specimens were exposed to the elements after the completion of the tests, resulting in reinforcement corrosion, the uniformity was calculated as reinforcement exposure.

3. Sample preparation

3.1. Selection and classification of samples

Performance-based earthquake engineering requires the identification of different damage states of structural members. Piers are one of the most important components of a bridge system and determine the overall performance of the bridge under seismic loading. Fig. 2 depicts the different bridge damages. The ability to predict the different damage states of bridge piers after an earthquake can be useful in restoring services or prescribing appropriate remedial measures. This paper studies the application of computer vision inspection methods to bridge structure inspection. The core of the method is image analysis processing. This work aims to develop a deep learning (DL) method capable of detecting damage at different scales in concrete structures and constructing an automatic model for identifying structural surface damage on bridges.

To use deep learning to make interpretive text, we must prepare many pairs of input images and the interpretive text that goes with them. Nevertheless, there is not yet a text dataset that describes how damaged high-speed railroad bridges are in this way. Then, we put together a dataset with pictures of real bridge damage. In particular, the dataset was made from photos the authors' team took between 2012 and 2017 while researching seismic performance tests of high-speed railroad bridges.

Common bridge structural damage manifestations are mainly: concrete members cracks, exposed reinforcement, spalling, honeycomb, pockmarks, and reinforcement corrosion. In this paper, we refer to the damage classification of different researchers to classify the damage level of high-speed railroad reinforced concrete bridges and build and train CNNs networks.

i) The common five-level scheme, namely: none, slight, moderate, extensive, and complete damage [31].

ii)The damage degree classification of component failure modes based on target detection and identification, i.e., fine cracks, wide cracks, concrete spalling, reinforcement exposure, and reinforcement



Fig. 3. HSR RC bridge pier damage photos and the bridge test site.. (a) Pseudo dynamic tests [36], (b) pseudo-static tests [37], (c) shaking table tests[38], and (d) full bridge tests [39,40].

Table 1
Convolution neural network parameters after fine-tuning.

Layer Number	Input Size			Operation Type	Feel the Wild				Output
	High	wide	thick		High	wide	thick	Step length	Output
1	227	227	3	Conv1	11	11	3	4	96
2	55	55	96	MaxPooling	3	3	96	2	96
3	27	27	96	Conv2	5	5	96	1	256
4	27	27	256	MaxPooling	3	3	256	2	256
5	13	13	256	Conv3	3	3	256	1	384
6	13	13	384	Conv4	3	3	384	1	384
7	13	13	384	Conv5	3	3	384	1	256
8	13	13	256	MaxPooling	3	3	256	2	256
9	6	6	256	FC6	–	–	–	–	4096
10	1	1	4096	FC7	–	–	–	–	2048
11	1	1	2048	FC8	–	–	–	–	4

Note: Conv is the convolutional layer, MaxPooling is the top pooling layer, and FC is the fully connected layer. The dropout layer, ReLU layer, and Norm layer are not listed.
A pseudocode of the proposed Alexnet algorithm.is summarized in Table 2. The full code is available in the Appendix A.

Table 2
Pseudocode of the proposed Alexnet algorithm.

Algorithm: Alexnet Algorithm

1 **Import torch**
i. Convolution operation, convolution kernel is 11x11x3, step size 4, the number of 96 convolution operations, the output 55 × 55 × 96, Relu after the result is still 55 × 55 × 9
ii. Activation function
iii. After Maxpool2d, the image data becomes 27 × 27 × 96; after a convolution operation with a convolution kernel of 5 × 5 × 96, a step size of 1, and a number of 256, the output is 27 × 27 × 256, and the result after Relu is still 27 × 27 × 256
iv. After Maxpool2d, the image data becomes 13 × 13 × 256; after a convolution operation with a convolution kernel of 3 × 3 × 256, a step size of 1, and a number of 384, the output is 13 × 13 × 384. Then, after another convolution operation, the image data remains unchanged,
v. After a convolution operation with a convolution kernel of 3 × 3 × 384, a step size of 1, and a number of 256, the output is 13 × 13 × 256. Then, after a nonlinear activation and a maximum pooling, the image data becomes 6 × 6 × 256.
vi. Adaptive pooling
vii. A spreading operation is performed on 6 × 6 × 256, followed by three full concatenation operations to obtain 1000 features.
2 **Main Functions**
3 **Input**
4 **Output**

buckling classification [32].

iii) Damage method classification considering different performance limit states, i.e., spalling, core crushing, and buckling of reinforced concrete piers under seismic loading [33].

iiii) Reinforcement corrosion is the main factor affecting the seismic performance of reinforced concrete columns [34].

iiiii) the ability to detect damage at different scales in concrete structures, including, i.e., microscopic and macroscopic cracks [35].

In this study, a well-structured dataset was collected based on experiments conducted by the authors' group in recent years. It is impossible to train various damages as a separate class considering the limited number of samples of various damage images. Therefore, this paper classifies bridge structure damage into three major categories according to the characteristics of each type of damage: cracks, corrosion, and defects, and collates a total of 360 images of each type of damage, including 120 images of cracks, corrosion, and defects. The samples were randomly taken by different digital cameras and belonged to different bridges, and the image sizes were above 10 megapixels. All photographs in this paper are from experiments done by the authors' research team [36–40]. Since the photographs used in this paper are based on the proposed static test of bridge piers, the proposed dynamic test of bridge piers, the shaking table test of bridge piers, and the shaking table test of the entire bridge, This paper selected 100 images from the three types of damage as training samples, and the remaining 20 were used for practical application tests. Fig. 3 shows the HSR RC bridge pier damage photos and the bridge test site (see Table 1, Table 2).

3.2. *Image cropping and training*

Artificial intelligence methods have been applied to detect concrete damage through image classification by CNNs, which can be classified into three types: image PATCH classification, bounding box regression, and semantic segmentation [41,42].

Image segmentation and bounding box regression have been widely used to detect cracks and other types of damage in infrastructure with good results [43,44]. It can be seen that image segmentation is one of the most popular techniques for analyzing and classifying photographs of

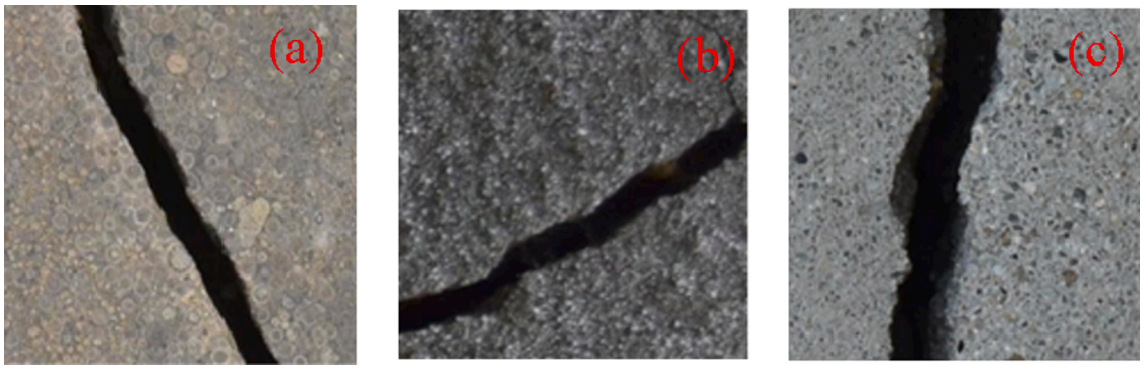


Fig. 4. Concrete crack c samples. (a) An ideal case of cracked samples, (b) cropped to one side of the crack, and (c) samples in which the crack almost entirely occupies the sub-image.

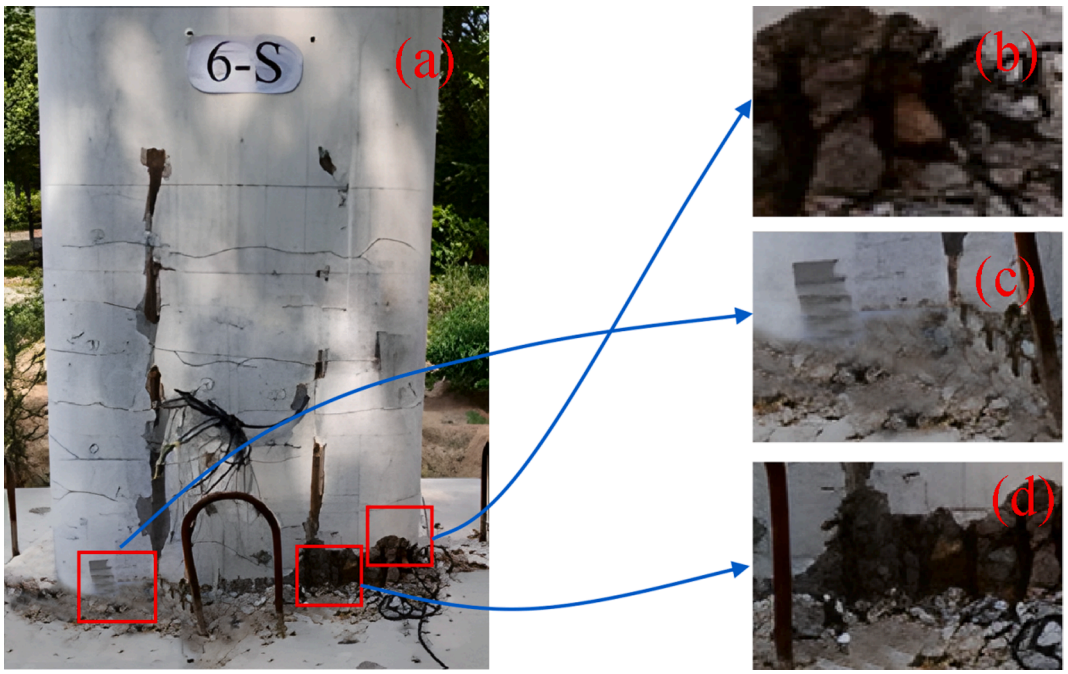


Fig. 5. Concrete spalling samples. (a) Experimental original image of the bridge pier; (b) edge of concrete spalling damage; (c) edge of bridge pier damage; and (d) center of concrete spalling damage.



Fig. 6. Sample of reinforcement corrosion. (a) Experimental original image of the bridge pier; (b) corrosion of exposed reinforcement; (c) concrete cracks at the base of the pier leading in exposed reinforcement; and (d) exposed reinforcement area at the base of the pier.

different objects.

The sample size is more than 10 megapixels, and the images contain more background information and are more complex. Therefore, the training images are randomly cropped into sub-images of size 227 × 227 to highlight the samples' single information and reduce the influence of irrelevant interference factors. Due to the scaling problem, the cropped image will have the following conditions:

Concrete crack sample (Fig. 4)

Fig. 4 (a) shows the sample of cracks in the ideal case; Fig. 4 (b) indicates the cropped to one side of the cracks only; Fig. 4 (c) displays the cracks almost occupying the sub-image due to scaling problems.

(2) Defective samples

The defective region generally occupies most of the image area, and a more typical defect and its cropped sub-image are shown in Fig. 5. Fig. 5 (b) shows the edges of the defect area, which are better distinguished; Fig. 5 (c) shows the structural angles that produce confusion; Fig. 5 (d) has a monotonous sub-image feature in the central region of the defect, which is similar to the background image.

(3) Reinforcement exposure

More typical rebar corrosion with its cropped sub-image is shown in Fig. 6.

For the above-cropped images produced in a series of situations, this paper selects three types of damage samples based on the following principles; ① Concrete crack, both sides of the cracks are clear; ② concrete spalling, the edges of the defective area are better distinguished; ③ reinforcement exposure. Typical damage photos are shown



(a) Concrete crack (b) Reinforcement exposure (c) Concrete spalling

Fig. 7. Typical structural disease images.

in Fig. 7.

From the cut damage samples, 1,300 samples of concrete cracks, concrete spalling, and reinforcement exposure were selected, and 2,700 samples of background, totaling 6,600 samples, were used as training samples.

3.3. Image pre-processing

Data pre-processing plays an important role in many deep learning algorithms. Many algorithms work best when the data is normalized and whitened. In order to reduce the impact on the image's overall brightness, the mean reduction process is applied to the samples.

$$x_1^+ = x_i - \overline{X} \tag{1}$$

where $\overline{X}$ is the mean of sample X;x_i is sample data (xi);x_i^+ is the sample after the mean reduction process.

4. Model training

4.1. Migration learning based on AlexNet CNNs

Convolutional neural networks are generally built from convolutional, pooling, fully connected, and classifier layers. The correct classification rate of CNNs increases with the depth and width of the network, as seen in the previous ImageNet competition winners [45]. Deep CNNs have multiple weights to be trained, and training a brand-new network is labor-intensive and generally requires many data samples to support. Therefore, it becomes preferred to migrate using classical CNNs after fine-tuning them in combination with migration learning [46].

Typically, migration learning refers to the process by which a model trained on one problem is used in some way on another related problem [35]. This technique is flexible because it directly uses pre-trained models for feature extraction, pre-processing, and integration into full-fledged new models. It reduces the training time of neural network models and can lead to lower generalization errors [47]. The context of transfer learning techniques for image classification is that if the model is trained on a large and general dataset, the model will effectively achieve a visual overview of the most important particular features or attributes [48,49].

Considering some objective conditions such as computer hardware, and data scarcity, a classical CNNs AlexNet [50] is chosen in this paper, and the last fully connected layer (FC8) is adjusted according to the requirements of the current task. The adjusted AlexNet is shown in Fig. 8. According to migration learning, the initial values of network weights are initially the weights obtained from AlexNet training on the ImageNet dataset [51].

The AlexNet model was selected as the migration target considering the hardware and other conditions. All the convolutional layers of the AlexNet model were locked during the training process, and only the last two fully connected layers were fine-tuned and trained.

4.2. Hyperparameter setting

The bridge structure damage image recognition model is trained using the caffe deep learning framework. The advantage of Adam is that after bias correction, the learning rate of each iteration has a definite range, which makes the parameters relatively smooth. The number of parameters of deep CNNs is large, and the computational demand is high, so the use of GPU can greatly improve computational efficiency. This paper uses NVIDIA GeForce GTX 960M to train the damaged samples. The training takes about 1.5 h, and the GPU hardware overhead is small and fast.

4.3. Training and validation results

In this part, the model's findings will be presented and discussed. All experiments were conducted using MATLAB 2019 on two computers: A Lenov P920 graphics workstation desktop, an Intel Xeon Gold 6230 16-core 2.1 GHz deep learning server GPU, 64G DDR4 memory, 512G solid state drive + 4 T3 3.5 in. wired hard disk, NVIDIA GeForce RTX 6000 graphics card. A DELL Precision T7920 graphics workstation desktop, 2 Intel Xeon Silver 4110 8-core 16-thread main frequency 3.8G deep learning simulation finite element analysis 3D modeling, 128G DDR4 memory, M.2 1 TB PCIe NVMe SSD + 4 TB 7.2 K RPM SATA 6Gbps 3.5-inch wired HDD; AMD Radeon ProW6600, 8 GB, GDDR6 graphics card. In the process of training the model, a validation set is needed to test the model's performance. In this paper, the training and validation sets are randomly selcected from 6600 training samples in a fixed ratio of 4:1. Considering that the training set and validation set are randomly selected from the training sample set, there are training sets and validation sets with different combinations of training samples. In order to obtain the best-performing model, five different combinations of training and validation sets are randomly selected from the same training samples to train five models, and the training results of the five models are shown in Table 3. The results show that different combinations of training and validation sets with the same sample set will have some effect on the training effect, but the effect is not significant. Among them, Model-2 performs best in its validation set, with an accuracy of 98.214 2 %. Fig. 9 (left) shows the training process of model 2. The highest correct rate of the Model validation set is 98.2443 % at 90 Epoch, and the correct rate of the training set is 100 %.

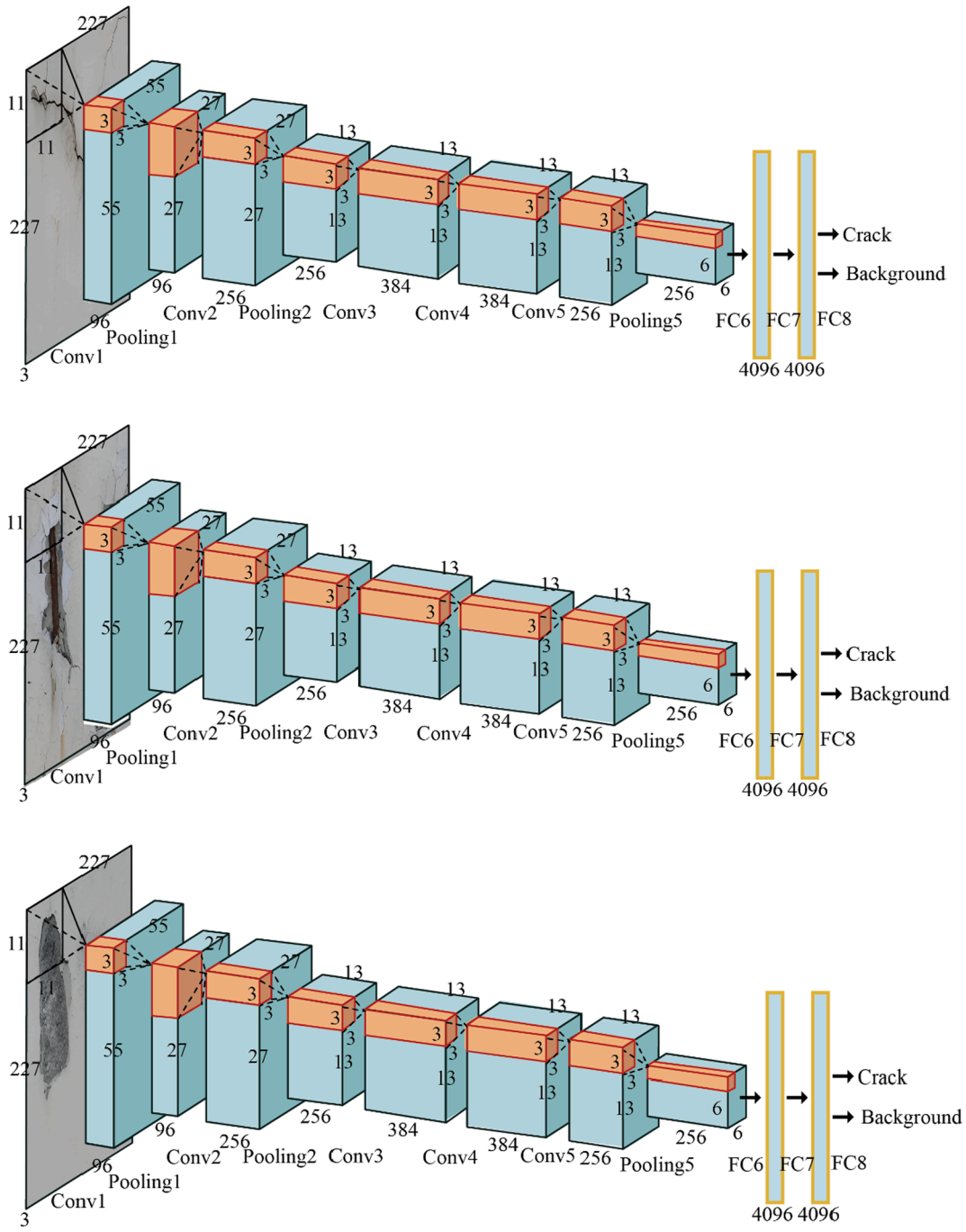


Fig. 8. Schematic architecture of the AlexNet after adjustment.

Table 3
Validate accuracy with different combinations of train and validate sets.

Model number	Validation sets the correct rate
Model-1	97.9911 %
Model-2	98.2443 %
Model-3	96.77 4 %
Model-4	96.1310 %
Model-5	98.1399 %

5. Practical applications

In order to meet the needs of engineering applications, the trained model is tested for its ability to generalize out-of-sample images. This paper divides the test into two parts: (i) the sample classification test after hand-picked characterization; (ii) the junction test in the actual engineering images.

The authors calculate and compare using the Unet and Unet + VGG16 models. After 50 iterations of training with the same hyper-parameters, the model converges on the area. Fig. 9 (right) depicts the decline function. As depicted in the figure, the loss function decreases rapidly as the number of iterations increases in the early training period, indicating that the model parameters converge toward the optimal solution. After approximately ten iterations, the loss function value decreases to between 0 and 0.1, and after 30 iterations, the loss function of the Res-Unet model gradually converges to between 0.04. The accuracy of the Unet and FCN + VGG16 models is 91.7 % and 92.2 %, respectively, as seen in Fig. 9.

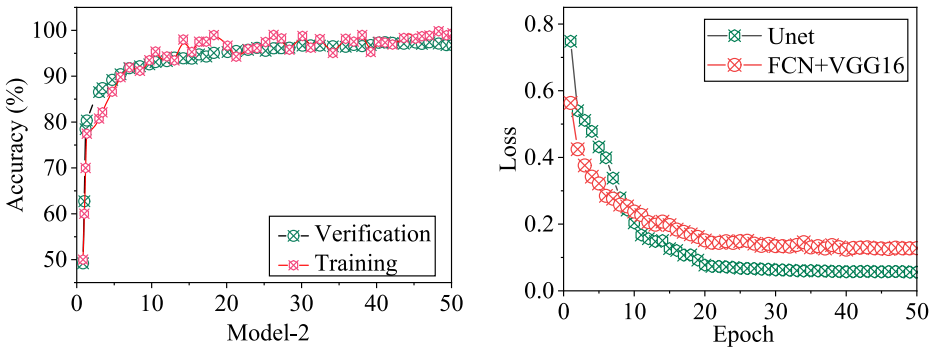


Fig. 9. Accuracy (left) and loss (right) during training and validation.

Table 4
Test results.

Test sample type	Number of tests	Identify the correct quantity	Number of recognition errors	Correct recognition rate
Concrete cracks	150	129	21	86.0 %
Reinforcement exposure	150	123	27	82.0 %
Concrete spalling	150	105	45	70.0 %
Background	150	147	3	98.0 %
Overall	600	504	96	84.0 %

True Label						Recall
	Concrete crack	129	12	7	2	84%
	Concrete spalling	16	105	17		69%
	Rebar exposure	4	28	123	1	73%
	Background	1	5	3	147	94%
	Accuracy	86.00%	70.00%	82.00%	98.00%	
		Concrete crack	Concrete spalling	Rebar exposure	Background	
		Predicted Label				

Fig. 10. Confusion matrix of test examples for three damage types.

5.1. *Manual qualitative post-sample classification test*

According to Section 1.2, the damaged images not added to the training were cropped, including 20 images of each type, totaling 60 images, from which test samples were selected, 150 images each of cracks, rust, defects, and background, totaling 600 images. The test results are shown in Table 4. The correct recognition rate of cracks and corrosion is relatively low, and that of defects is high.

A confusion matrix, which illustrates the discrepancy between real and anticipated values, may be used to examine the performance of a deep learning model in more depth. The confusion matrix is a table with the horizontal coordinate True Label and the vertical coordinate Predicted Label, where diagonal entries represent injury cases that are correctly detected based on global percentages, non-diagonal entries are used for error detection, and the sum of all values is 100 %. Ideally, all non-diagonal values are zero.

Fig. 10 illustrates the confusion matrix for the test instances of the three damage categories. The horizontal and vertical axes show the predicted and true forms of damage. Based on the confusion matrix, true positives (TP), true negatives (TN), false positives (FP), and false

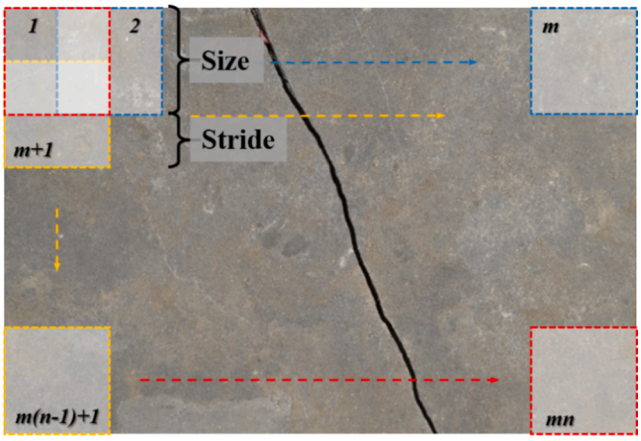


Fig. 11. Sliding Window.

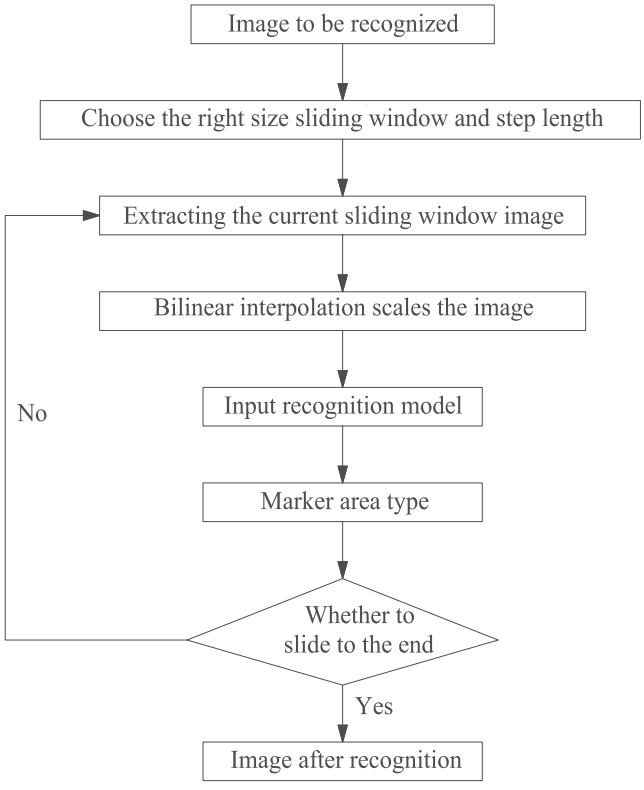


Fig. 12. Chart of disease recognition.

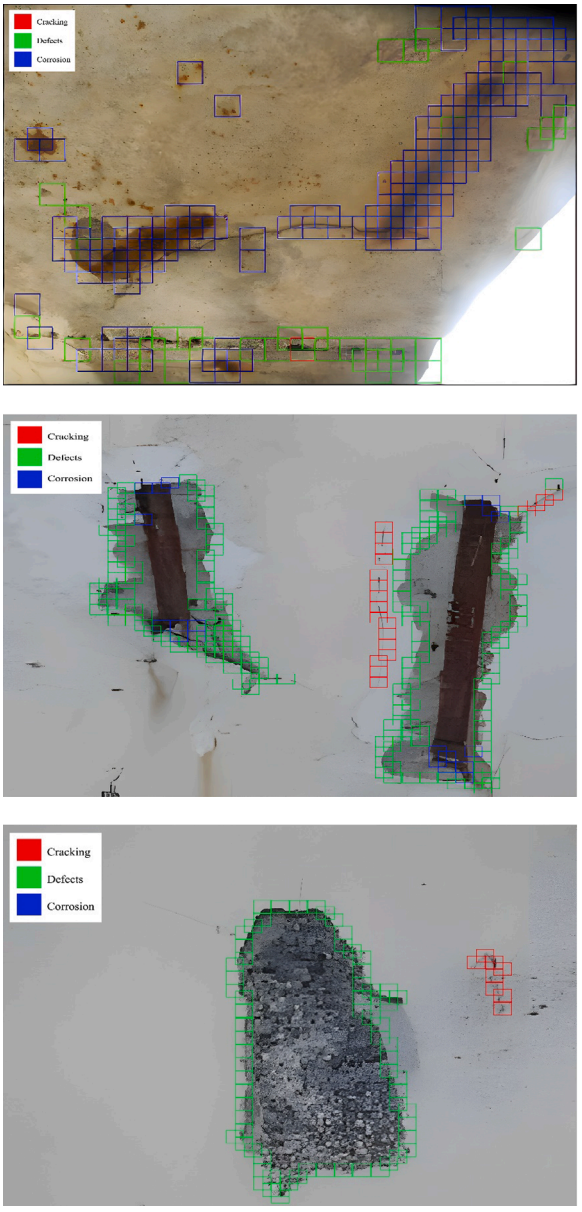


Fig. 13. Bridge damage disease recognition. (a) Damage recognition photo 1, (b) damage recognition photo 2, and (c) damage recognition photo 3.

negatives (FN) may be determined for each kind of damage, as well as several assessment measures.

$$\text{Accuracy} = (\text{TP} + \text{TN})/(\text{TP} + \text{TN} + \text{FP} + \text{FN}) \quad (2)$$

$$\text{Rcall} = (\text{TP})/(\text{TP} + \text{FN}) \quad (3)$$

Fig. 10 demonstrates that the accuracy and recall for the three evaluated damage types are, respectively, 86 %, 70 %, 82 %, and 84 %, 69 %. 73 %. The results indicate that Concrete fracture and Rebar exposure was recognized at a greater accuracy rate. It should be noted that the photos detected are not from the training or validation collections. The accuracy rate is somewhat lower than the photos in the training and validation sets.

5.2. Damage recognition in images

In practice, the acquired images are of high pixel size, and the information within the images is complex and diverse. Therefore, it is necessary to introduce a sliding window to scan large-size images for structural damage identification and localization. As shown in Fig. 11, a sliding window (size) of a certain size slides over the image area in a certain step (stride), and the type of damage in the window area is identified at each slide.

In this paper, we set stride = size /2 to avoid the undesirable samples in Section 1. 2 and overlay the scanned area to prevent missing the damaged area. Due to the effect of image scaling at the time of capture, it is necessary to use different scale sliding windows (Size) to scan the images. The model only accepts 227 × 227 size images, and the different size sliding windows need to be scaled by bilinear interpolation to fit the model input requirements, the specific recognition process is shown in Fig. 12, and the partial bridge damage image recognition is shown in Fig. 13.

The model can identify and locate the bridge structure damage from the practical application test well. However, there are many recognition errors, mainly in two aspects: (i) the surrounding environment is mixed in the images, which may affect the recognition; (ii) there is a big difference between the damaged images to be recognized and the training samples.

Here, the issue of "huge discrepancies between the illness pictures to be detected and the training samples" is addressed. "random selection from the training sample set into the training and validation sets" is not the sole component of the authors' research. It also involves "proving the capacity of the trained model to generalize to non-sample photos for engineering applications," which is this study apart from previous studies on the same subject.

Consider Fig. 14, for instance; the FP indicates the number of sub-images with the background that were misidentified as cracks; FN indicates the number of sub-images with cracks that were misidentified as the background. Fig. 14′ s left side depicts the original picture, while the

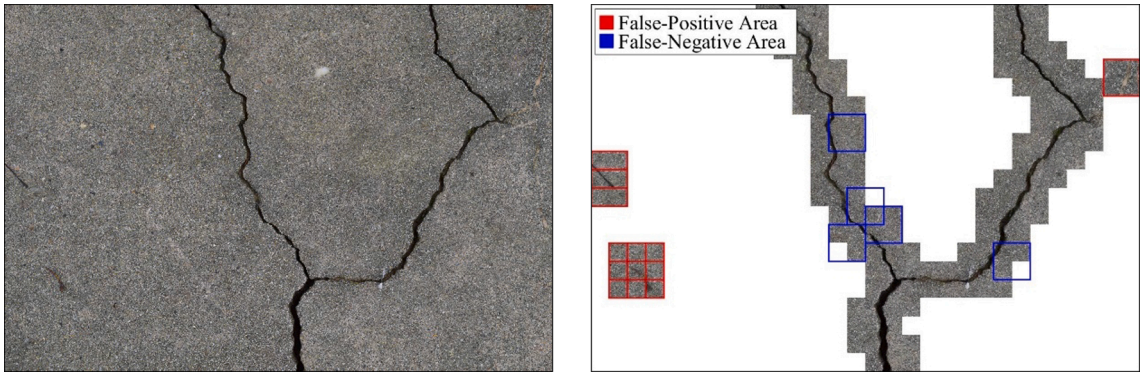


Fig. 14. Test image recognition.

right side depicts the outcome of processing. Analysis of the original picture reveals that the backdrop is rough, with strong noise and crack-like target interference. The model is durable and highly adaptable to inhomogeneous lighting and noise interference. The research demonstrates that the recognition accuracy of damage images for practical engineering applications is somewhat lower than the recognition accuracy of the theoretical analysis. However, the present state of the art still reaches a high level of accuracy, as shown in Fig. 14.

6. Conclusion

In this paper, we study the application of deep CNNs to classify and recognize bridge structure damage images and construct an automatic recognition model for three types of bridge structure damage using migration learning techniques. In this paper, we summarize the characteristics of bridge structure damage images and classify bridge damage into three major categories, and select damage samples based on certain principles. In the training process, the best model is selected by adjusting the training and validation sets and observing the training performance of the model. Finally, the model is tested in practical engineering applications, the correct recognition rate is determined on the untrained samples, and a multi-scale sliding window is combined to propose a damage recognition localization method. The test results show that the model has a high recognition rate and can be used to identify structural damage to bridges rapidly.

This study involves random selection from the training sample set, training set, and validation set. It also validates the final training model's capacity to generalize to non-sample pictures for engineering applications. This aspect of the work sets it apart from previous research in the same study area.

This study attempted to calculate the Unet model and compare it to the Unet + VGG16 model. After training the model for 50 iterations with the same hyperparameters, it converged on the area. Researchers found that both the Unet and FCN + VGG16 models performed well, with the former having a 91.7 % accuracy and the latter a 92.3 % accuracy.

CRediT authorship contribution statement

Lingkun Chen: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing – original draft, Writing – review & editing, Visualization. **Wenxin Chen:** Investigation, Writing – original draft, Writing – review & editing, Visualization. **Lu Wang:** Software, Investigation, Writing – review & editing. **Chencheng Zhai:** Investigation, Writing – original draft, Writing – review & editing. **Xiaolun Hu:** Software, Writing – review & editing. **Linlin Sun:** Investigation, Data curation, Writing – review & editing. **Yuan Tian:** Data curation, Investigation, Writing – original draft, Writing – review & editing, Visualization. **Xiaoming Huang:** Investigation, Writing – original draft, Writing – review & editing, Visualization. **Lizhong Jiang:** Investigation, Writing – original draft, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Compliance with Ethical Standards

This article does not contain any studies with human participants performed by any authors.

Data availability statement

Some or all data, models, or codes that support the findings of this study are available from the corresponding author upon reasonable request.

Appendix A

(*continued on next page*)

(*continued*)

```
# Alexnet Algorithm
import torch
from torch import nn
class Net(nn. Module):
    def __init__(self):
        super(Net, self).__init__()
        # Convolution operation, convolution kernel is 11x11x3, step size 4, the number of 96 convolution operations, the output 55 x 55 x 96, Relu after the result is still 55 x 55 x 9
        self.features = nn.S.Sequential(
            nn.Conv2d(3, 96, kernel_size=11, stride=4, padding=2),
            nn.ReLU(inplace=True), #activation function
            #After Maxpool2d, the image data becomes 27×27×96; after a convolution operation with a convolution kernel of 5×5×96, a step size of 1, and a number of 256, the output is 27×27×256, and the result after Relu is still 27×27×256
            nn.MaxPool2d(3, 2),
            nn.Conv2d(96, 256, 5, 1, 2),
            nn.ReLU(inplace=True),
            #After Maxpool2d, the image data becomes 13×13×256; after a convolution operation with a convolution kernel of 3×3×256, a step size of 1, and a number of 384, the output is 13×13×384. Then, after another convolution operation, the image data remains unchanged
            nn.MaxPool2d(3, 2),
            #After a convolution operation with a convolution kernel of 3×3×384, a step size of 1, and a number of 256, the output is 13×13×256. Then, after a nonlinear activation and a maximum pooling, the image data becomes 6×6×256
            nn.Conv2d(256, 384, 3, 1, 1),
            nn.Conv2d(384, 384, 3, 1, 1),
            nn.Conv2d(384, 256, 3, 1, 1),
            nn.ReLU(inplace=True),
            nn.MaxPool2d(3, 2)
        )
        #Adaptive pooling
        self.avavfpool = nn.AdaptiveAvgPool2d((6, 6))
        #A spreading operation is performed on 6×6×256, followed by three full concatenation operations to obtain 1000 features.
        self.classifier = nn.Sequential(
            nn.Flatten(),
            nn.Linear(9216, 4096),
            nn.ReLU(inplace=True),
            nn.Dropout(),
            nn.Linear(4096, 1000)
        )
    def forward(self, x):
        x = self.features(x)
        x = self.avfpool(x)
        x = self.classifier(x)
        return x
#    Main Functions
model = Net()
print(model)
# Input
input = torch.ones((64, 3, 224, 224)) # Input image is 224×224×3
#Output
output = model(input)
print(output.shape)
```

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